

Scheduling five processes

P1–P5 arrive within a few time units of each other. This works out the Gantt chart, turnaround time and waiting time for five scheduling policies.

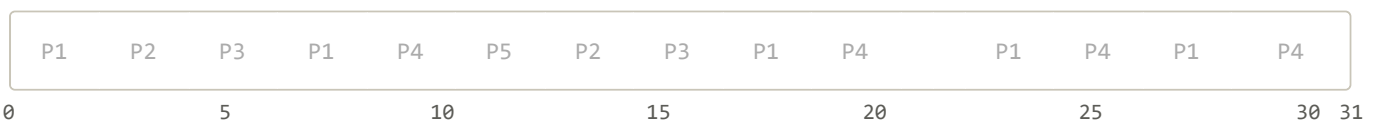
Turnaround time = completion – arrival. Waiting time = turnaround – burst.

Process data

Process	Arrival	Burst	Priority
P1	0	10	3
P2	1	5	1 (lowest)
P3	2	4	5 (highest)
P4	3	9	2
P5	3	3	4

P1 P2 P3 P4 P5

1. Round robin (quantum = 2)



Process	Arrival	Burst	Completion	Turnaround	Waiting
P1	0	10	28	28	18
P2	1	5	22	21	16
P3	2	4	16	14	10
P4	3	9	31	28	19
P5	3	3	21	18	15

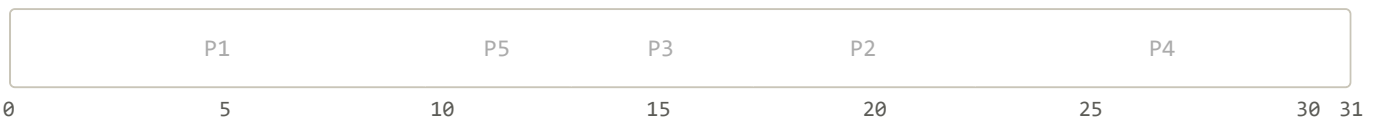
Average turnaround

Average waiting

21.8

15.6

2. Shortest job first



Process	Arrival	Burst	Completion	Turnaround	Waiting
P1	0	10	10	10	0
P2	1	5	22	21	16
P3	2	4	17	15	11
P4	3	9	31	28	19
P5	3	3	13	10	7

Average turnaround

16.8

Average waiting

10.6

3. Priority scheduling, preemptive



Process	Arrival	Burst	Completion	Turnaround	Waiting
P1	0	10	17	17	7
P2	1	5	31	30	25
P3	2	4	6	4	0
P4	3	9	26	23	14
P5	3	3	9	6	3

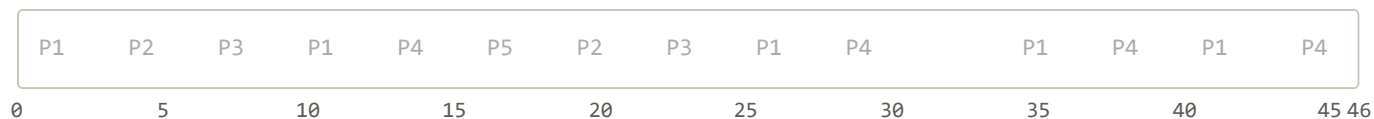
Average turnaround

16.0

Average waiting

9.8

4. Round robin (quantum = 2, switch overhead = 1)



Gray blocks are the 1-unit context-switch overhead inserted before each dispatch.

Process	Arrival	Burst	Completion	Turnaround	Waiting
P1	0	10	42	42	32
P2	1	5	33	32	27
P3	2	4	23	21	17
P4	3	9	46	43	34
P5	3	3	31	28	25

Average turnaround

33.2

Average waiting

27.0

5. First come, first served



Process	Arrival	Burst	Completion	Turnaround	Waiting
P1	0	10	10	10	0
P2	1	5	15	14	9
P3	2	4	19	17	13
P4	3	9	28	25	16
P5	3	3	31	28	25

Average turnaround

18.8

Average waiting

12.6

Summary

Algorithm	Average turnaround	Average waiting
Round robin (q=2)	21.8	15.6

Algorithm	Average turnaround	Average waiting
Shortest job first	16.8	10.6
Priority, preemptive	16.0	9.8
Round robin (q=2, overhead=1)	33.2	27.0
First come, first served	18.8	12.6

Priority scheduling wins here because P3, the shortest job, also holds the highest priority and runs almost immediately. The overhead variant of round robin shows how much throughput a small quantum loses once each switch has a fixed cost.